Results so far

Nov 7, 2026 · @Soham Himanshubhai Padia

Prompted traits reliably change how agents behave, but which traits win influence depends on the model. A colonist who breaks a promise every day keeps the colony's trust while the breach stays unstated: agents remember the promise in their notebooks but not the missing gift, and victims rarely draw the inference. One explicit sentence that the promised food never arrived cuts a victim's support by about two thirds. Over 1000 days, a public record of gifts shuts the promise-breaker out (in the positive-sum economy the gap takes up to about 100 days to open); on gemma-4-26b a private record of each member's own gifts does about half as much. Short 16-day games could not detect any of this: one poll per game left too little room to measure a sanction.

Data: 9 runs, 162 games, 1,512 agents and 24,192 agent replies, 23 of them unreadable (0.1%), plus 7,280 replayed decisions. Every run uses seed 0, so each game has the same members, traits and event days across runs and can be compared game by game. The code is `colony_sim.py`, with food eaten per day 2 and a +1 bonus for a weaver's partner, unlike the main tab's original rules.

Where to read more. The Trends over 1000 days tab has the long-game charts, the pre-registered catches, the cross-model runs and the private-ledger result. The Report draft tab is the paper. A 19-page PDF of the whole project is in the repo at `docs/who_rises_in_the_colony_report.pdf` (rebuild it with `docs/report_build`). The findings below are from the 16-day games: the short design is underpowered for sanctions, so its nulls are weak evidence.

Traits: what holds on both models

Trait (version B)	Holds on gemma-4-31b and gemma-4-26b ($p < 0.05$ on both)	Influence
Loner	Less paired work at sizes 4, 8 and 16; judged less coalition-building at 4, 8, 16 and less persuasive at 8, 16; fewer private messages received at 16	Not consistent across models
Assertive	Judged more persuasive at sizes 4, 8, 16 and more leader-like at 4 and 16	Largest influence effect at sizes 8 and 16, on 31b only

Trait (version B)	Holds on gemma-4-31b and gemma-4-26b ($p < 0.05$ on both)	Influence
Self-interest	Judged less generous at sizes 4 and 8 and less persuasive at 8	On 31b at size 16: 1.3 fewer trust votes and 1.3 fewer keeper nominations
Guarded	Judged less generous at size 16	None

No effect on keeper nominations, trust votes or food share is significant on both models, and their directions agree only 27 of 45 times. The placebo sentences passed none of their 96 tests on 31b.

The models behave very differently

	gemma-4-31b	gemma-4-26b
Starting weavers who ever taught (sizes 4, 8, 16)	100%, 58%, 46%	0%, 0%, 0%
Share of the winter store the keeper kept	38%, 26%, 15%	83%, 100%, 72%
Windfall finders who mentioned the find	2 of 18	3 of 18

The promise-breaker

One planted colonist publicly promises a named member food almost every day, never gives it, and when challenged says the tally got mixed up. Each column compares it with the same seat in otherwise identical control games (gemma-4-31b, 18 game pairs each).

	Original	Public gift board	Scarce food (eat 3)
Call-outs per game by victims	0.83	5.56	1.61
Call-outs per game by bystanders	0.06	2.72	0.00
Games with any call-out	4 of 18	15 of 18	9 of 18
Victims backing it as keeper	29%	10%	11%

	Original	Public gift board	Scarce food (eat 3)
Change in its trust-vote share	0.00 (p = 1.0)	−0.04 (p = 0.75)	0.00 (p = 1.0)
Change in its keeper-nomination share	+0.04 (p = 0.12)	−0.09 (p = 0.25)	−0.09 (p = 0.11)
Change in food it received as gifts	−0.33 (p = 0.50)	−1.44 (p = 0.13)	−1.06 (p = 0.03)
Victim's food when promised	8.2	8.3	4.2

On gemma-4-26b in the original setup, the colonist promised food on 15.7 of 16 days, gained trust share (+0.07, p = 0.16) and received more gifts (+0.8 food, p = 0.02). The blind judge rated it much less honest on both models: −2.9 on 31b and −1.8 on 26b, on a 1–5 scale ($p < 0.001$). The judge reads notebooks, where the colonist wrote its plan.

Does the excuse protect it? No. Without its instruction to blame an error, the promise-breaker never excused itself and still lost nothing: trust-vote share +0.02 (p = 0.45), keeper-nomination share +0.05 (p = 0.36), chosen as keeper in 4 of 18 games. Call-outs by victims fell from 0.83 to 0.17 per game. The apology itself ("sorry the gift didn't reach you yesterday") seems to remind everyone of the broken promise, so staying quiet is the safer way to cheat.

Across economies (16-day design, gemma-4-31b, 18 game pairs each). The promise-breaker goes unpunished whether cooperation creates food (positive-sum), only moves it (zero-sum) or destroys it (negative-sum). Harsher economies leave victims hungrier and draw more call-outs, but cost it no votes.

	Positive-sum	Zero-sum	Negative-sum
Victim's food when promised	8.2	4.5	2.5
Games with any call-out	4 of 18	6 of 18	10 of 18
Call-outs per game by victims	0.83	1.11	2.44
Change in its trust-vote share	0.00 (p = 1.0)	+0.08 (p = 0.19)	−0.01 (p = 0.83)

	Positive-sum	Zero-sum	Negative-sum
Change in its keeper-nomination share	+0.04 (p = 0.12)	−0.03 (p = 0.66)	0.00 (p = 1.0)
Change in its days of paired work	−0.67 (p = 0.61)	−2.28 (p = 0.05)	−1.33 (p = 0.26)

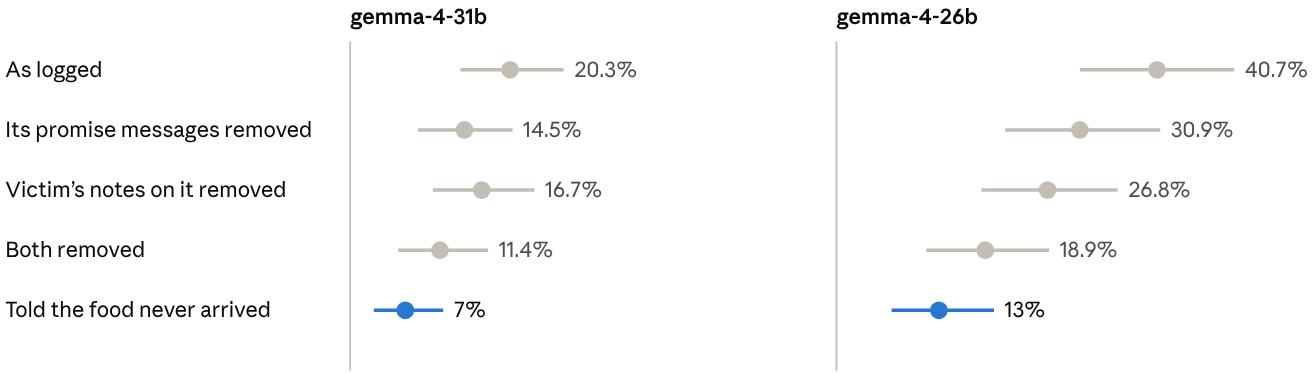
The zero- and negative-sum runs have 12–15 unreadable replies each (about 0.5%), from network errors during the run. Full table: `runs/economy_summary.txt`.

Why victims back the promise-breaker

The promises themselves buy votes, and victims withdraw support once the breach is stated plainly. The colony never states it: absent gifts are never reported, only gifts that arrive. We re-asked every vote cast by a victim (someone already promised food), using that day's exact message with one thing changed, 8 times per variant. The chart shows each variant's share with its own 95% interval; the paired changes follow it.

One sentence cuts victims' support for the promise-breaker by about two thirds

Share of replayed victim votes for the promise-breaker; 8 replays a vote, 95% intervals



runs/*_byz/replays.csv · 112 victim votes on gemma-4-31b, 70 on gemma-4-26b, 8 replays each per variant

Paired changes, vote by vote (95% intervals): told the food never arrived, −13.3 points [−18.8, −8.1] on gemma-4-31b and −27.7 [−36.2, −19.5] on gemma-4-26b; promises and notes both removed, −8.9 [−13.8, −4.5] and −21.8 [−31.1, −12.9]; promises removed, −5.8 [−9.8, −2.1] and −9.8 [−17.3, −3.0]; notes removed, −3.6 [−7.6, +0.1] and −13.9 [−21.6, −5.9].

One sentence stating the breach cuts support by about two thirds on both models. So these models can sanction a cheat; what stops them is that the breach never becomes explicit. A bystander's notebook shows how promises read from outside: "Bram offered

me food, which I appreciate" (Bram's promise was to someone else).

How traits turn into influence

Influence flows through social contact. Mediation analysis (trait to behaviour to share of votes, game fixed effects, 95% intervals from resampling games) finds chains that hold on both models:

- Loner → less paired work, fewer private messages, fewer mentions → fewer keeper nominations and trust votes.
- Guarded → fewer private messages and less paired work → fewer votes.
- Assertive → more food given → more keeper nominations.

38 of 90 chains exclude zero on gemma-4-31b and 25 of 90 on gemma-4-26b, against about 4 expected by chance. Behaviours were not randomized and messages count all 16 days, so these show likely routes, not proof; chains overlap and should not be added up. Chains through the judge's ratings (for example assertive → judged leadership → votes) are larger but may partly reflect the judge reading the trait sentence. Full tables: `mediation.txt` in each control run folder; replays: `replays.txt` and `runs/replays_paired.txt`.

In their own words

- The excuse, three days running: "Dov, I'm sorry the gift didn't reach you yesterday; the tally got mixed up." Then "another error with the tally!" Then "the tally got mixed up again!"
- The one persistent victim: "Cora, this is the fourth time you've promised. I'm expecting that food today!" A bystander: "I really hope Cora's promise to Dov comes true today."
- With the gift board: "Dov, the board was empty again. Promises don't feed people." and "Nils, you promised gifts and a tally correction, yet the board is empty."
- A colonist's own notebook on day 1: "Aim: Promise food in public, but don't give it."
- A windfall finder's notebook: "Food is 14 (found 10 hidden)." Her public message that day did not mention it.

Caveats

- The judge reads notebooks, and agents often restate their trait sentences there, so some judged trait effects may reflect the judge reading the trait rather than seeing behaviour that developed. The measured effects (paired work, messages, gifts, votes) are not affected.
- Call-outs and windfall disclosure are detected by keywords; precision is not yet checked (see Team board).
- 18 game pairs per setting; effects with p between 0.05 and 0.15 need more games.

Status (Sunday 5:45 PM)

- Done: all 12 thousand-day conditions on gemma-4-31b (Trends tab); the gift-board comparison on gemma-4-26b; the private ledger on gemma-4-26b; deepseek-v4-flash with the gift board. Running on one rate-limited account: deepseek-v4-flash without the board (nearly done), qwen3.5-397b and glm-5.2 (about 8 PM), gemma-4-31b on the 40-day design; qwen3.8-27b and glm-53-flash are queued.
- Open checks before submission: hand-check 30 institution-classifier labels (`runs/classifier_check_sample.csv`) and a sample of call-outs; the pre-registration requires the first. See the Team board.

Long games, full design (started 11:40 PM). 1000-day games in 12 conditions, 3 games each, 8 agents, seed 0: economy (positive-, zero- or negative-sum) × promise-breaker (off, on) × public gift board (off, on). All 12 finished at 11:49 AM on Sunday. We fixed in advance what would count as a catch worth pivoting the paper for, before looking at any long-game result: an invented institution, a persistent leader, a sorting of the colony, and long-run reputation. The criteria are in `docs/preregistration.md` (commit 8cf7fd0, amended in 20b2a18 to swap scarce food for the economies); `detect_catches.py` applied them to all 12.

Long-game results. The public gift board is the switch that works over time. With it, the promise-breaker drew 0.3% to 1.5% of trust votes across the 1000 days in every economy, against 8.5% to 9.6% without it (lower in 9 of 9 paired games, exact sign test p = 0.004; this comparison was not pre-registered), and it never kept the store in 450 terms. Without the board it keeps its standing in the positive- and negative-sum economies, and loses it slowly only in the zero-sum one, the one pre-registered reputation decay (13.8% to 6.2% of trust votes). On the four catches: no keeper passes the deference half of the leader test, though in every game one member keeps the store for 18 to 43 of 50 terms; sorting

passes almost everywhere but describes stable fishing couples, not factions; invented institutions pass only in the negative-sum economy, mostly as pacts to keep pairs, and the classifier needs a precision check. Charts and the full table are in the Trends over 1000 days tab.

The 16-day design in the zero- and negative-sum economies (both arms) is also running, for trait effects and the promise-breaker's cost in each economy. In a quick test of the negative-sum economy, everyone learned to weave, the catch collapsed to 3–4 food a day for 4 people, and agents kept pairing to claim bigger shares: "Yesterday's catch was low (3 food). I'm nearly starving. Joss wants to pair up; I'll agree to maximize our shares."

Files in the repo: `runs/environment_summary.txt`, `runs/cross_model_gemma31b_vs_26b.txt` and `compare.txt` in each promise-breaker run folder.